Freight GHG emissions report

Organisation	Example Freight A/S
Reporting period	1 Jan 2026 to 31 Mar 2026
Level	Transport or hub services (GLEC Framework v3.3, Section 2, Chapter 1)
GHG Protocol	Scope 3, category 4 (upstream transportation and distribution)
Emission factors	desnz-2026+nl-iww-2026+eu-mrv-2024+glec-3.3
Calculated	10 November 2026, 18:16 UTC

Calculated with Sootcase following the GLEC Framework v3.3 and ISO 14083:2023 methodology. Conformity with ISO 14083 is not declared by this report.

Key figures

Total emissions (WTW)
4.74
t CO2e

Transport, operational (TTW)
3.27
t CO2e

Transport, energy provision (WTT)
0.76
t CO2e

Hubs (WTW)
0.71
t CO2e

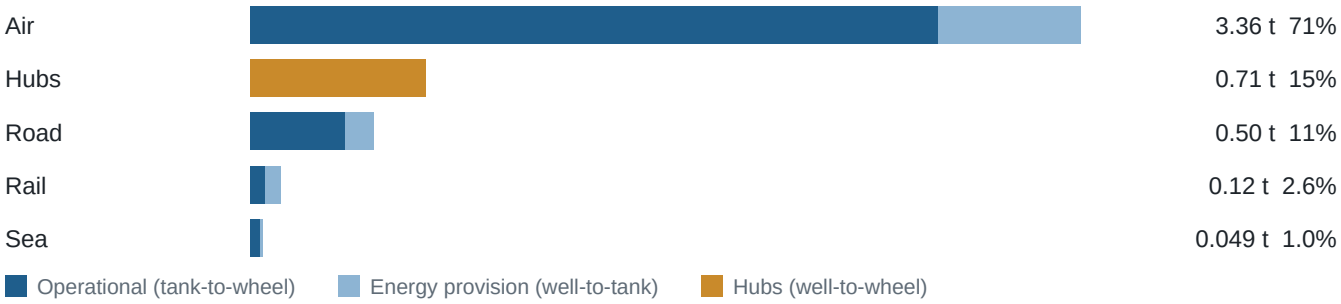
Transport activity
19,683
tonne-km

Transport intensity (WTW)
204.7
g CO2e / tonne-km

Shipments calculated
3 of 4
1 failed

Cargo
35.5
tonnes

Emissions by mode of transport



Mode	Legs	Tonne-km	TTW kg CO2e	WTT kg CO2e	WTW kg CO2e	g CO2e/tkm	Share of transport
Air	1	2,379	2,783	578.1	3,361	1,413.0	83%
Road	1	3,984	382.5	115.5	498.0	125.0	12%
Rail	1	6,600	59.2	62.0	121.2	18.4	3.0%
Sea	1	6,720	40.6	8.74	49.3	7.3	1.2%
Transport	**4**	**19,683**	**3,265**	**764.3**	**4,030**	**204.7**	**100%**

Emissions at hubs

Hub type	Stops	Tonnes handled	WTW kg CO2e	g CO2e/t
Warehouse	1	20.0	710.0	35,500.0
All hubs	**1**		**710.0**	

Emissions by energy carrier

Energy carrier	TTW kg CO2e	WTT kg CO2e	WTW kg CO2e	Share of transport
Not stated (published intensity or load model)	3,265	764.3	4,030	100%

Fuels are named by the fuel factor used for measured fuel; electricity by the grid it was drawn from.

GHG Protocol scopes

For the company buying the transport

Scope	kg CO2e
Scope 3, category 4 (upstream transportation and distribution)	4,740

For the operator of the vehicles and hubs

Scope	kg CO2e
Scope 1: fuel burnt in the vehicles	3,265
Scope 2: electricity used by the vehicles	0
Scope 3, category 3: producing and delivering the fuel and energy	764.3
Hubs (well-to-wheel, not split by scope)	710.0

Distances

Share of tonne-km by the type of distance used (ISO 14083 asks for it to be stated).

	Shortest feasible	Great circle	Actual	Not stated
Air		100%		
Road				100%
Rail				100%
Sea				100%
All transport		**12%**		**88%**

Data quality

Primary data 0.0% Modelled 0.0% Default values 100%

Shares of total well-to-wheel emissions. Primary: measured fuel or the operator's own values. Modelled: from shipment inputs such as load or energy use. Default: published emission intensities.

Sources of default values

Source	WTW kg CO2e	Share
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Table 1, air: aircraft type unknown, short-haul (< 1500 km), including the +95 km DAF, for use with the great circle distance	3,361	71%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Table 3, logistics hubs (GILA project: Fraunhofer IML, Politecnico di Milano, GreenRouter; median values): warehouse, ambient (n = 92)	710.0	15%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Europe and South America road, Table 8: rigid truck 26-32 t GVW, average/mixed load, diesel; includes GLEC's +5% distance conversion, so it is per shortest feasible distance	498.0	11%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, rail, Europe: EU average where the traction is unknown, 18.4 g CO2e/t-km WTW (p. 98, based on 62% of EU track being electrified); the TTW/WTT split weights GLEC's Table 4 diesel and Table 5 electric average/mixed values 38% / 62%; per shortest feasible distance (GLEC Section 1 Chapter 4, Rail, p. 46)	121.2	2.6%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Tables 18-19, container vessels (Clean Cargo trade lane factors, reporting year 2025): industry average (origin-destination pair unknown), dry, 13 + 60.4 g CO2e/TEU-km (WTT + TTW) / 10 t per TEU (GLEC's average when the cargo mass is not known); includes GLEC's +15% distance conversion, so it is per shortest feasible distance	49.3	1.0%

Largest shipments (3 of 3)

Shipment	Route	Tonnes	Tonne-km	WTW kg CO2e	g/tkm	Share
A2	Copenhagen Kastrup Airport (CPH) -> Frankfurt Main Airport (FRA) (air)	3.50	2,379	3,361	1,413.0	71%
A3	rail	20.00	6,600	831.2	18.4	18%
A1	road, sea	12.00	10,704	547.3	51.1	12%

Coverage

Shipments in the input	4
Calculated and included	3
Failed and excluded	1
Transport legs	4
Hub stops	1
Cargo, tonnes	35.5

Why shipments failed

Reason	Shipments
line #: weight_kg: "abc" is not a number	1

Failed shipments (1 of 1)

Shipment	Error
BAD	line 7: weight_kg: "abc" is not a number

Notes on omissions and deviations

- 1 of 4 shipments could not be calculated and are excluded from all totals (BAD). Reasons: 1 × line #: weight_kg: "abc" is not a number.
- Hub emissions are reported well-to-wheel only: the hub factors used carry no split into operational and energy provision emissions. Operational (TTW) and energy provision (WTT) totals therefore cover transport only.
- For 87.9% of tonne-km the type of distance was not stated (distances given without distance_type).
- Tonne-km combine more than one type of distance; see the shares per mode.
- Default emission intensities are used where no primary data (measured fuel or operation category values) was supplied for a leg or hub; their sources are listed.

Method and supporting information

Per-shipment results with factors, sources and distances: Sootcase batch output for the same input. Method: https://github.com/majidkabir/sootcase

Emissions are calculated per leg as tonne-km × emission intensity (× a distance adjustment factor where the distance type differs from the factor's), or from measured fuel or energy where given, and per hub stop as tonnes (or containers) handled × the hub's emission intensity, following the GLEC Framework v3.3 and ISO 14083:2023. Every leg's distance, method, data type and factor source are in Sootcase's per-shipment results for the same input. GLEC Framework default values: Smart Freight Centre, used with permission; no affiliation or endorsement is implied.

These are calculations for your own reporting. Sootcase is not certified against ISO 14083; conformity depends on how the data was collected and the boundaries chosen.